KU LEUVEN

FACULTY OF ENGINEERING SCIENCE

MASTER OF ARTIFICIAL INTELLIGENCE

Robotics

Homework A

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1 Q1

As Fig. 1 shows, one 3-DOF positioning arm can exhibit one, two, or infinitely many inverse position solutions, depending on the target point.

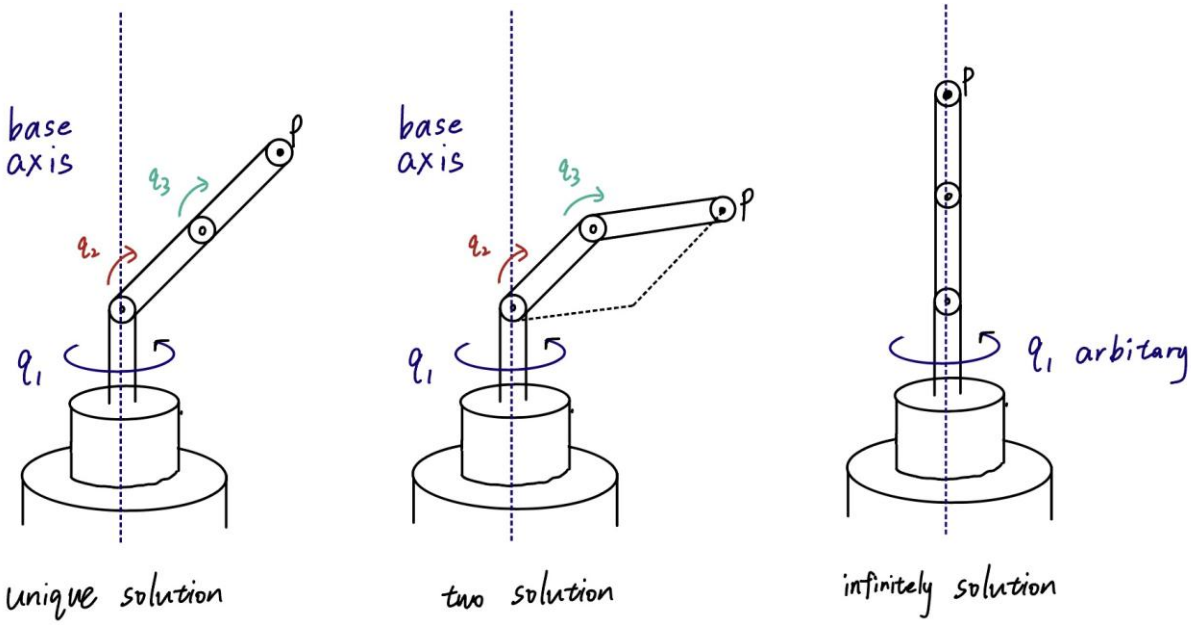

Fig. 1. Three inverse position kinematics cases for a 3-DOF positioning arm with base yaw q_1, shoulder pitch q_2, and elbow pitch q_3.

In the left panel, the target lies on the workspace boundary. The arm is fully stretched, so the elbow-up and elbow-down branches merge into one singular configuration. This gives one inverse position solution.

In the middle panel, the target lies at a generic reachable point. The base yaw stays unique, and the planar two-link geometry gives the standard elbow-up and elbow-down configurations. This gives exactly two inverse position solutions.

In the right panel, the target lies on the base rotation axis. The base yaw does not change the end-effector position in that case. The variable q_1 is free, so infinitely many inverse position solutions exist.

2 Q2

I define the configuration of the rolling coin as

$$q = (x, y, \phi, \theta) \in \mathbb{R}^2 \times \mathbb{T}^2.$$

Here (x, y) is the position of the contact point, ϕ is the heading, and θ is the rolling angle. The lecture models this system with two non-holonomic Pfaffian constraints. These constraints do not reduce the dimension of the configuration space. They only reduce the dimension of the feasible instantaneous velocity set.

I use a constructive three-step algorithm.

First, I move the coin to the target position. I turn in place until the heading points to the goal. Then I roll straight until the coin reaches (x_g, y_g).

Second, I set the desired heading. I turn in place until the heading becomes ϕ_g.

Third, I correct the rolling angle without changing the position or the heading. I execute the closed loop

1. roll forward by L,
2. turn in place by π,
3. roll forward again by L,
4. turn in place again by π.

This loop returns the coin to the same position and the same heading. It still changes the rolling angle by the total rolled distance $2L$. I choose L so that the final value equals θ_g modulo 2π.

This construction reaches any goal configuration $(x_g, y_g, \phi_g, \theta_g)$ from any initial configuration. The system is instantaneously constrained, but it is globally maneuverable.

3 Q3

The given rotation matrix is

$$R = \begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

I use

$$\mathrm{tr}(R) = 1 + 2\cos\theta.$$

Since $\mathrm{tr}(R) = 0$, I obtain

$$\cos\theta = -\frac{1}{2}, \qquad \theta = \frac{2\pi}{3}.$$

Next, I compute

$$[\widehat{\omega}] = \frac{1}{2\sin\theta}(R - R^T).$$

Because $\sin(2\pi/3) = \sqrt{3}/2$, this gives

$$[\widehat{\omega}] = \frac{1}{\sqrt{3}}\begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Matching this skew-symmetric matrix with the standard form yields

$$\omega = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{bmatrix}.$$

Therefore, one valid set of exponential coordinates is

$$\omega\theta = \begin{bmatrix} \frac{2\pi}{3\sqrt{3}} \\ -\frac{2\pi}{3\sqrt{3}} \\ -\frac{2\pi}{3\sqrt{3}} \end{bmatrix}.$$

The pair $(-\omega, -\theta)$ represents the same rotation, so equivalent answers also exist.

4 Q4

I use the complete planner suggested by the lecture: a visibility graph in C_{free}, followed by A* search on that graph.

As Fig. 2 shows, the point robot moves from the start to the goal through the visible obstacle vertices. For polygonal obstacles and a point robot, this construction is complete.

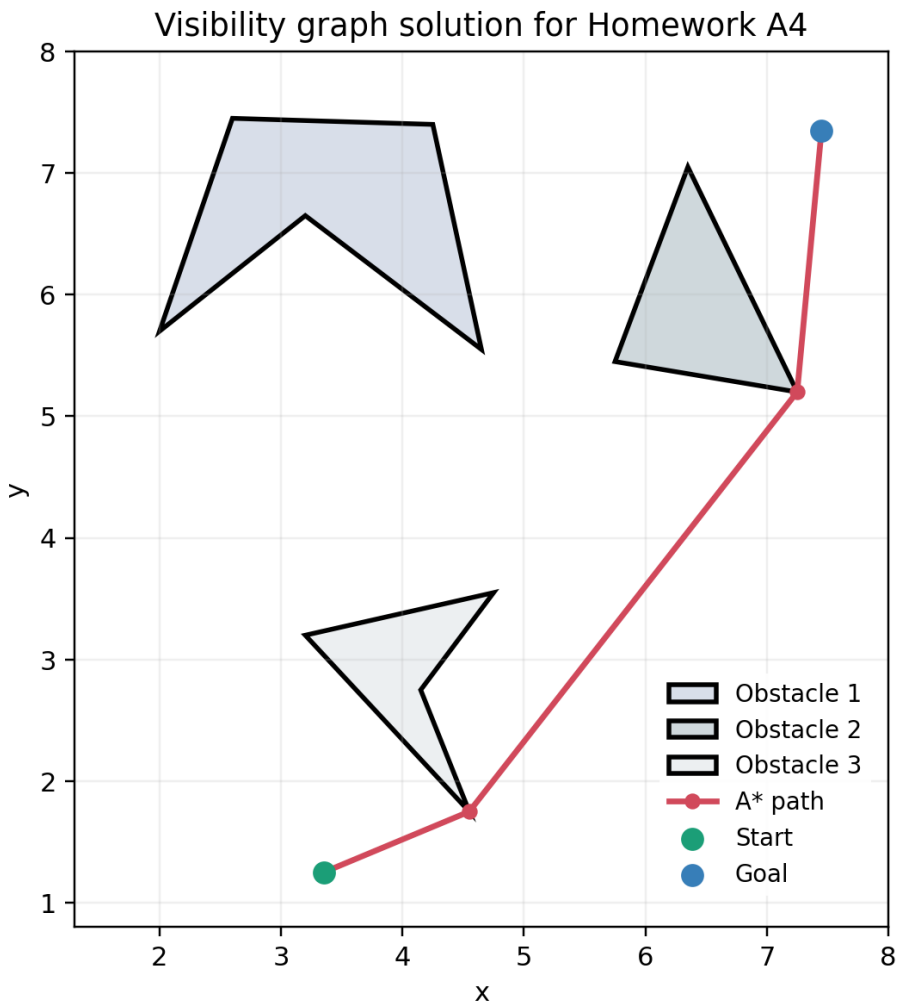

Fig. 2. Visibility-graph path for the point robot in the workspace of Homework A4.

I approximate the obstacle vertices as follows:

Obstacle 1:
[(2.0, 5.7), (2.6, 7.45), (4.25, 7.4), (4.65, 5.55), (3.2, 6.65)]

Obstacle 2:
[(5.75, 5.45), (6.35, 7.05), (7.25, 5.2)]

Obstacle 3:
[(3.2, 3.2), (4.75, 3.55), (4.15, 2.75), (4.55, 1.75)]

I set

start = (3.35, 1.25)
goal = (7.45, 7.35)

The planner returns the path

(3.35, 1.25)
(4.55, 1.75)

(7.25, 5.20)

(7.45, 7.35)

The total path length is 7.840207.